

Mixed-Cation Validation Report

Editable local report test

The leading formulation remains FA0.90MA0.10.

CHOSE Centre for Hybrid and Organic Solar Energy | Matteo Ginesi | 2026-08-03

Keywords: perovskite, mixed-cation, validation

BEST PCE 21.28% S08	MEAN PCE 19.79% 8 samples	STABILITY 95% best retained	MEAN VOC 1.09 V cohort	OPEN ISSUES 3 quality review
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EDITABLE REPORT IDENTITY

Title

Mixed-Cation Validation Report

Author / laboratory

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Approval state

Pending researcher signature

Executive summary

The leading formulation remains FA0.90MA0.10.

PROJECT PRJ-2026-014	PIPELINE chose	EVIDENCE 28 files 18 measurements 7 findings
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Research context

Objectives

Compare MA/FA ratios and identify the most stable high-efficiency device stack.

Methodology

Versioned preparation, mapped JV data and deterministic comparison.

QUALITY REVIEW: 1 error | 2 warnings | causal interpretation remains blocked until provenance is complete

Solution composition | SOL-B04

COMPONENT	FUNCTION	QUANTITY	COMPOSITION
DMF	Primary solvent	1.60 mL	80% v/v
DMSO	Co-solvent	0.40 mL	20% v/v
FAI	A-site solute	365.3 mg	90 mol%
MAI	A-site solute	39.7 mg	10 mol%
PbI2	Lead halide	1152.5 mg	1.00 eq

Device stack | STK-003/v2

05	Au	Back contact	80 nm
04	Spiro-OMeTAD	Hole transport	180 nm
03	FA/MA perovskite	Photoactive absorber	540 nm
02	SnO2	Electron transport	32 nm
01	Glass / FTO	Substrate + front contact	2.2 mm

Experiment coverage

EXP-041	S01-S03	100 C / 30 min	6 measurements	Reviewed
EXP-052	S04-S05	105 C / 25 min	4 measurements	Reviewed
EXP-067	S06-S08	100 / unit missing	24 imported	Review

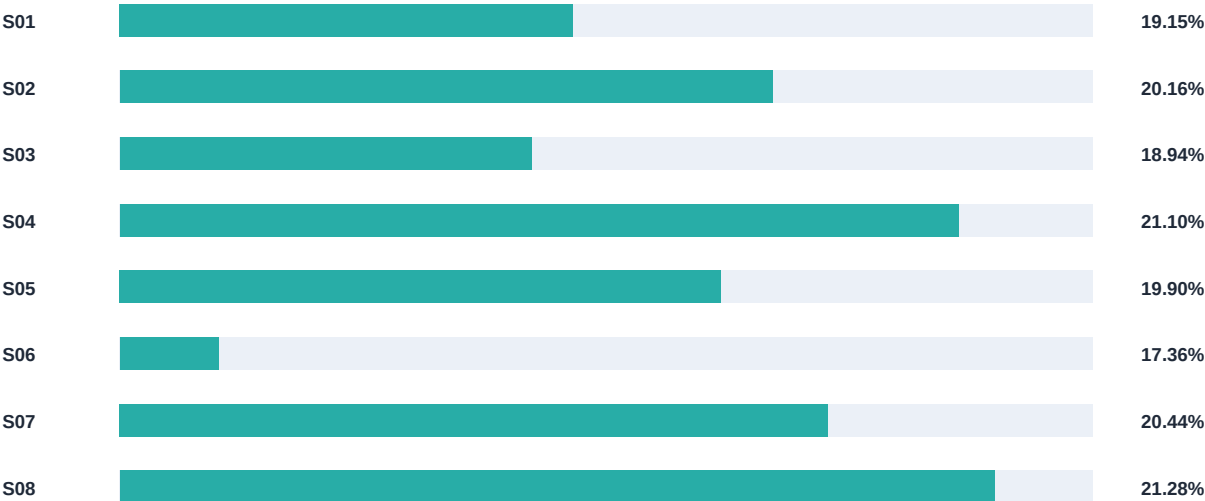
Complete JV result table

Every displayed source-aligned value remains editable in this exported copy.

SAMPLE	FORMULATION	BATCH	VOC	JSC	FF	PCE	STABILITY	HYSTERESIS
S01	FA0.85MA0.15	B01	1.08	22.7	78.1	19.15	89	3.2
S02	FA0.85MA0.15	B01	1.1	23.2	79	20.16	91	2.8
S03	FA0.80MA0.20	B02	1.07	22.9	77.3	18.94	85	4.1
S04	FA0.90MA0.10	B03	1.12	23.5	80.2	21.1	94	2.1
S05	FA0.90MA0.10	B03	1.09	23	79.4	19.9	92	2.5
S06	FA0.75MA0.25	B04	1.05	21.8	75.8	17.36	78	5.9
S07	FA0.85MA0.15	B05	1.11	23.1	79.7	20.44	90	2.7
S08	FA0.90MA0.10	B06	1.13	23.4	80.5	21.28	95	1.9

PCE comparison

All samples | export-time deterministic snapshot



MEAN PCE 19.79%	MEAN VOC 1.09 V	MEAN FF 78.8%	BEST STABILITY 95%
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S08 remains the current performance leader.

Evidence-linked findings

Simulated AI remains advisory and separate from researcher-authored conclusions.

96	FA0.90MA0.10 is the strongest candidate Evidence: S04, S08 7 metrics Status: accepted
91	S06 is a multi-metric outlier Evidence: S06 IQR + robust z Status: review
88	Batch effect is smaller than formulation effect Evidence: Grouped comparison Status: accepted
84	Stability and hysteresis are inversely associated CUSTOM AUTHOR SECTION Additional notes Evidence: Spearman preview Status: review The author requested one additional validation run.

EDITABLE RESEARCHER DECISION

Conclusions	The current cohort supports validation but not causal inference.
S04 and S08 proceed to validation; S06 remains under review.	

Limitations
Small demonstration cohort; no causal inference.

Approval / signature state
Pending researcher signature

PROVENANCE CLASSES

RAW	Local source-aligned measurements	CALCULATED	Deterministic KPI and comparisons
RESEARCHER	Objectives, conclusions and approval	AI	Simulated advice requiring review